MD ARIFUR RAHMAN

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No Sponsorship Required

MACHINE LEARNING ENGINEER | OPERATIONS RESEARCH | DATA SCIENTIST

Building ML models, optimization systems, and data-driven decision tools that improve operational performance.

PROFESSIONAL SUMMARY

Applied machine learning engineer and operations researcher with experience across mathematical optimization (LP/MIP, RL, HMM), machine learning, and data engineering. Builds and deploys ML models — including forecasting, anomaly detection, and LLM-based systems — and optimization models for scheduling, routing, and resource planning, running scenario/sensitivity analysis to support decision-making. Collects, validates, and analyzes large datasets to surface trends and translate findings into actionable recommendations. Proficient in Python, SQL, R, and cloud platforms (AWS, GCP, Azure). Published researcher (10 papers, 260+ citations) with a track record of delivering projects from concept through production deployment.

AREAS OF EXPERTISE

Machine Learning & AI • Supervised & Unsupervised Learning • Unsupervised Clustering • Reinforcement Learning • Generative AI & LLMs • Mixed Integer Programming (MIP) • Network Optimization, Scheduling & Routing • Scenario & Sensitivity Analysis • MLOps & Model Deployment • Forecasting & Predictive Analytics • Anomaly Detection • Statistical Analysis & Hypothesis Testing • Data Engineering & Pipeline Development • Cloud Platforms (AWS, GCP, Azure) • Data Visualization & Dashboards • Cross-Functional Collaboration • Research & Technical Publication

TECHNICAL SKILLS

Programming & Data:	Python | SQL | R | C++ | MATLAB | Git | Java
ML & Modeling:	TensorFlow | PyTorch | scikit-learn | Hugging Face | LLMs (GPT, BERT) | LLM Fine-Tuning (PEFT/LoRA/QLoRA) | RAG (LangChain, LangGraph) | Computer Vision (YOLOv8) | AutoML | Hyperparameter Tuning
Optimization & OR:	Linear Programming (LP) | Mixed Integer Programming (MIP) | Reinforcement Learning | Hidden Markov Models | Gurobi | CPLEX | SCIP | OR-Tools | Multi-Objective Optimization | ML Surrogate Modeling
Cloud & Data Engineering:	AWS | Google Cloud (Vertex AI, BigQuery) | Azure | Docker | ETL / Data Pipelines
Reporting & Visualization:	Pandas | Power BI | Tableau | Alteryx

PROFESSIONAL EXPERIENCE

PARTNERSHIP FOR INNOVATION (GEORGIA TECH), ATLANTA, GA

AI/ML Research Engineer — PIN Fellow **07/2025 – Present**

Applied optimization research under the Georgia-AIM grant — developing intelligent systems that close the loop between sensing, state estimation, and autonomous decision-making for industrial asset performance.

- Designed and calibrated a reinforcement learning + Hidden Markov Model optimization pipeline for real-time process control — RL learns the optimal action policy to maximize material utilization and minimize process failures.
- Led multi-objective optimization research using ML surrogate models to simultaneously minimize cost, maximize throughput, and satisfy operational constraints across competing design variables.
- Built a production RAG-based knowledge assistant (LangChain, FAISS, Docker) deployed on Hugging Face Spaces, and fine-tuned/calibrated multiple LLMs (BERT, RoBERTa, LoRA Mistral-7B via PEFT/QLoRA).
- Developed real-time anomaly detection and condition monitoring on high-frequency sensor data, enabling proactive, data-driven decision-making and reducing unplanned downtime.

NORFOLK SOUTHERN CORPORATION, ATLANTA, GA

Supervisor Associate – Mechanical Maintenance, Operations Division **01/2024 – 03/2025**

Built data pipelines and decision-support dashboards for railcar condition reporting and asset management across the rail network.

- Built complex datasets from multiple internal data sources; developed automated Alteryx/SQL reporting pipelines and GIS-enabled Power BI/Tableau dashboards for real-time asset health monitoring — adopted company-wide across all operational teams.
- Coordinated with Class I railroad carriers on capacity coordination and vendor contract management, translating data-driven insights into operational decisions that minimized equipment idle time.
- Led process improvement initiatives across multiple network locations; improved inspection turnaround time by 5% through workflow bottleneck analysis and optimization.

GEORGIA SOUTHERN UNIVERSITY (LANDTIE RESEARCH LAB), STATESBORO, GA

Research Assistant — ML & Data Analytics **08/2022 – 12/2023**

ML research on rail safety and reliability in collaboration with MxV Rail using live fiber-optic sensor datasets from active rail operations.

- Built CNN-LSTM and GRU anomaly detection models on live fiber-optic distributed acoustic sensing (DAS) datasets — 97% and 94% detection accuracy — demonstrating end-to-end data engineering and model development on large sensor datasets.

- Collaborated directly with MxV Rail on research programs for condition monitoring and structural health monitoring.

Prior Experience: Mechanical & Process Development Engineer, Aramit Limited (02/2018–08/2022) | Mechanical Maintenance Engineer, Walton Hi-Tech Industries (05/2017–01/2018) | Assistant Engineer – Mechanical Maintenance, KSRM Steel Plant (11/2015–04/2017)

SELECTED PROJECTS

- AI-Powered Last-Mile Delivery Optimization — Capacitated VRP with time windows solved via Gurobi (exact MIP) and OR-Tools (production-scale); XGBoost demand forecasting and Random Forest travel-time prediction; Streamlit dashboard with scenario/sensitivity analysis. https://github.com/arifme071/ai-last-mile-delivery-optimization, live: last-mile-delivery-optimization
- Warehouse Visual Intelligence — 5-agent CrewAI pipeline (YOLOv8 + GCP Vision API) for real-time safety monitoring; FastAPI backend, Streamlit dashboard, CI/CD on GCP. https://github.com/arifme071/warehouse-visual-intelligence , live: warehouse VI

EDUCATION | PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

GEORGIA INSTITUTE OF TECHNOLOGY, ATLANTA, GA

Master of Science in Computer Science (OMSCS) — part time, In progress

GEORGIA SOUTHERN UNIVERSITY, STATESBORO, GA

Master of Science in Applied Engineering (Major: Manufacturing Engineering) —(2022–2024)

CHITTAGONG UNIVERSITY OF ENGINEERING AND TECHNOLOGY, CHITTAGONG, BANGLADESH

Bachelor of Science in Mechanical Engineering (2010- 2014)

Certifications: Google Cloud Data Analytics Certificate | Coursera Operations Research Certification (National Taiwan University) | Alteryx Designer Core Certification | Coursera Applied Machine Learning (University of Michigan)

Professional Development: IBM Deep Learning & Neural Networks with Keras, Generative AI & LLMs | Google Generative AI Leader

Affiliations: ASME Member | ASNT Member | Gene Haas Foundation Scholar (2022–23)

RESEARCH & PUBLICATIONS

10 publications/submissions, 260+ citations across Elsevier, Springer, SPIE, and ASME — including reinforcement-learning-based process optimization and ML-based anomaly detection for large-scale sensor systems. ORCID 0009-0004-8766-5351